

Lab Task 1: Creating Activities and Navigating Between Them

Source Code:

- AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.ActivityApp">

        <activity
            android:name=".SecondActivity"
            android:exported="false" />

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity 1: activity_main.xml & MainActivity.java

- activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:gravity="center"
android:background="#B0F8E6"
android:orientation="vertical">

<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Go to second Activity" />

</LinearLayout>
```

- MainActivity.java

```
package com.demo.activityapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnNext;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btn);

        btnNext.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });
    }
}
```

Activity 2: Creating Another Activity: & activity_second.xml & SecondActivity.java

- activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:background="#D5BDED"
    android:orientation="vertical">

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Main Activity" />

</LinearLayout>
```

- SecondActivity.java

```
package com.demo.activityapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    Button bck;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_second);
        bck = findViewById(R.id.btn1);

        bck.setOnClickListener(v -> {
            Intent intent = new Intent(SecondActivity.this, MainActivity.class);
            startActivity(intent);
        });
    }
}
```

```
});  
}  
}
```

Lab Task 2: Creating and Using an Android Service

Source Code:

- AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<manifest xmlns:android="http://schemas.android.com/apk/res/android"  
  xmlns:tools="http://schemas.android.com/tools">  
  
  <application  
    android:allowBackup="true"  
    android:dataExtractionRules="@xml/data_extraction_rules"  
    android:fullBackupContent="@xml/backup_rules"  
    android:icon="@mipmap/ic_launcher"  
    android:label="@string/app_name"  
    android:roundIcon="@mipmap/ic_launcher_round"  
    android:supportsRtl="true"  
    android:theme="@style/Theme.ServiceApplication">  
  
    <service  
      android:name=".MyService"  
      android:exported="false" />  
  
    <activity  
      android:name=".MainActivity"  
      android:exported="true">  
      <intent-filter>  
        <action android:name="android.intent.action.MAIN" />  
  
        <category android:name="android.intent.category.LAUNCHER" />  
      </intent-filter>  
    </activity>  
  </application>  
</manifest>
```

- activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Service" />

    <Button
        android:id="@+id/btnStop"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Stop Service" />

</LinearLayout>
```

- MainActivity.java

```
package com.demo.serviceapplication;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnStartService, btnStopService;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        btnStartService = findViewById(R.id.btnStart);
```

```
btnStopService = findViewById(R.id.btnStop);

btnStartService.setOnClickListener(v -> {
    Intent intent = new Intent(MainActivity.this, MyService.class);
    startService(intent);
});

btnStopService.setOnClickListener(v -> {
    Intent intent = new Intent(MainActivity.this, MyService.class);
    stopService(intent);
});
}
```

Creating Service Class: **MyService.java**

```
package com.demo.serviceapplication;

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.widget.Toast;

import androidx.annotation.Nullable;

public class MyService extends Service {

    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show();
        return START_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show();
    }

    @Nullable
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```